

Simulating Just-in-Time Adaptive Interventions: A Config-Driven, LLM-Bootstrapped Framework

William Dennis

University of Bristol & National University of Singapore

Physical inactivity is a leading modifiable health risk [1]; just-in-time adaptive interventions (JITAI) deliver personalised prompts through wearables, but trials cannot probe counterfactuals and transition data is rarely public [2]. We reduce policy design to a config-driven pipeline for two simulators: LLM-bootstrapped patient models, evaluated against an optimal dynamic-programming bound and bandit baselines.

1. Motivation and approach

We formalise the intervention as a Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$ and build a framework in which *everything* is configuration: state spaces, action sets, transition models, rewards, and agents are declared in YAML, so experiments differ by config file alone (configs and tables in the repository) [3, 4].

2. LLM-bootstrapped transitions

The transition function $P(s' | s, a)$ is the hardest component of any patient model: it must encode how context, burden, and a delivered suggestion change behaviour, and it is rarely available as data. Rather than hand-curating tables or waiting for trials, we prompt an LLM (DeepSeek V4 Flash, with a GLM 5.2 comparison) to sample next-state distributions given a persona description, current state, and candidate action — a text world model [5]. Unlike data-driven simulators [2], which require logged trial data, bootstrapping needs no logged data — though it still requires hand-authored personas, states, and rewards in config. Personas were hand-curated by domain practitioners, loosely grounded in the behaviour-change literature: not arbitrary, but not validated against real patient data either. Park et al. [6] report agents reproducing held-out General Social Survey responses at 85% of participants’ own two-week response-replication consistency; we make no such population-replication claim for ours. Six sweeps of 22,320 prompts each (133,920 prompts, 10 samples per outcome) produced 31 JSON tables: \$24.52 (DeepSeek) + \$25.45 (GLM 5.2). Personas produced distinct sampled transition distributions, and the config’s Bayesian mechanism models burden as lowering future activity probability [2, 7].

3. Validation

On the Phase-1 persona tables, a HeartSteps-V2-style UCB (upper confidence bound) bandit reaches 47–95.8% of the optimal dynamic-programming bound across personas (50 seeds; 95% confidence intervals ± 0.6 –9.5pp); the idle policy alone reaches 96.8–100.1%, the values above 100% being Monte Carlo noise (± 3 –5) — the bound is essentially the idle policy — interventions underperform idle on these tables, a property of these personas’ bootstrapped tables, not a claim about JITAI efficacy in general. All baselines are internal to the bootstrapped tables (mock-on-mock; no ground truth; burden sensitivity untested). Eight literature-review reports document this background; the one quantitative comparison (a V2-style agent at 46.7% vs the paper’s 78.4%, on different data) did not reproduce.

4. PEARL-matched extension (Phase 2)

In the final two weeks we compared against the PEARL randomised controlled trial (RCT) [8], whose reinforcement-learning (RL) arm gained +295.9 steps over control ($p=0.0002$): we used 6 COM-B themes $\times$ 2 delivery times [9], plus idle — with an ϵ -greedy bandit and a COM-B barrier-weighted fixed arm [10–12]. The 14,040-prompt bootstrap (108 states $\times$ 13 actions $\times$ 10 samples) passed 11 of 17 checks; the six failures (inflated effect size $d=1.525$; fixed $>$ RL inversion; no attenuation; no burden saturation; inverted weekend; degenerate control t-test) trace to an idle-vs-action structure. A reward-penalty probe ($R = \text{step_reward} - \lambda \mathbf{1}[\text{action} \neq \text{idle}]$, $\lambda = 0.05$, $\text{step_reward} - 1/0/+1$; an anti-spam term from the Phase-1 reward design, independent of the PEARL comparison) does not restore the RL-vs-fixed ordering: 50-seed runs sit within noise (± 5 –7) and the ordering flips with the penalty (unpenalised RL 3.0 $>$ fixed 2.4; penalised fixed $-0.6 >$ RL -0.7) — the penalty taxes activeness. Calibration metrics, not trial outcomes; the probe is not a validated fix.

5. Conclusion

Policy design reduces to a cheap, config-driven pipeline, albeit mock-on-mock only, with a documented map of where the simulation diverges from published trials.

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